

ATS + AI SaaS Platform – Backend Explanation

1. Project Overview

This project is a backend system for an AI-powered Applicant Tracking System (ATS). It allows recruiters to post jobs, candidates to apply with resumes, and the system automatically analyzes resumes using NLP to generate a matching score.

2. Backend Architecture

The backend follows a clean layered architecture: Routes handle API endpoints, Controllers contain business logic, Models represent MongoDB collections, Services handle AI processing, and Utilities manage helper tasks like PDF parsing.

3. Authentication & Roles

JWT-based authentication is used. Users have roles: Candidate, Recruiter, and Admin. Role-based middleware ensures only authorized users can access specific APIs.

4. Multi-Tenant SaaS Design

Each recruiter belongs to a company. Jobs and applications are linked to a companyId. This ensures data isolation between different companies using the same platform.

5. Job & Application Flow

Recruiters create jobs. Candidates browse jobs and apply by uploading a resume. Each application is stored separately with resume path and status, forming a real hiring pipeline.

6. Resume Upload System

Multer is used to upload PDF resumes. Files are stored locally in an uploads folder, and only the file path is saved in the database.

7. AI Resume Parsing & NLP Matching

When a candidate applies, the resume PDF is converted into plain text using pdf-parse. The job description text and resume text are then compared using NLP techniques.

NLP Matching Logic:

1. Resume text and job description text are treated as documents.
2. TF-IDF (Term Frequency – Inverse Document Frequency) is applied to identify important words.
3. Common keywords between resume and job description are weighted.

4. A similarity score is calculated and normalized to a 0–100 range.
5. Higher score means better match.

8. Asynchronous AI Processing

AI analysis runs in the background. The application is saved immediately, and resume scoring happens asynchronously to keep APIs fast and scalable.

9. Application Status & Analytics

Recruiters can update application status (shortlisted, interview, rejected). Analytics APIs provide recruiter-level and admin-level insights using MongoDB aggregations.

10. Final Summary

This backend demonstrates real-world system design, SaaS multi-tenancy, AI integration using NLP, and scalable API architecture. It is production-ready and suitable for technical interviews.